

EECS 16A Designing Information Devices and Systems I

Homework 6

Submission Format

Your homework submission should consist of **one** file.

- `hw6.pdf`: A single PDF file that contains all of your answers (any handwritten answers should be scanned)

Submit the file to the appropriate assignment on Gradescope.

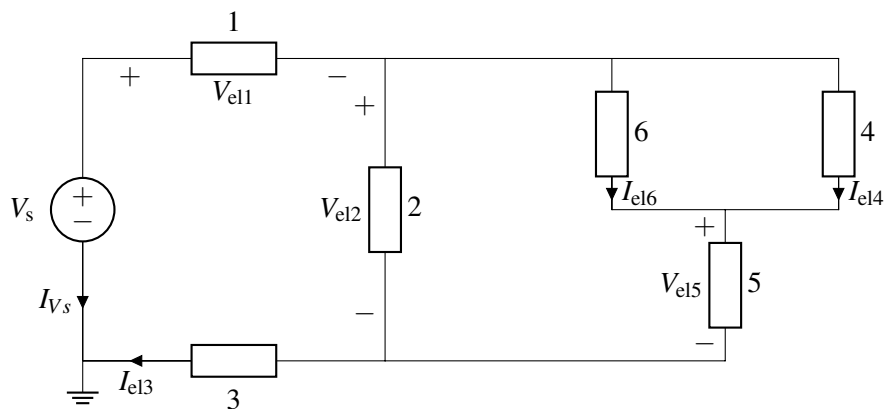
1. Reading Assignment

For this homework, please read Notes 11 through 13. Note 11 introduces the basics of circuit analysis. Note 12 will provide an overview of voltage dividers and resistors. Note 13 will refresh you on how simple 1-D resistive touchscreens work, as well as the notion of power in electric circuits.

- What is the value of having a systematic procedure for solving circuits?
- Describe the key ideas behind how the 1D touchscreen works. In general, why is it useful to be able to convert a “physical” quantity like the position of your finger to an electronic signal (i.e. voltage)?

2. Intro to Circuits

(Contributors: Panagiotis Zarkos)



- How many nodes does the above circuit have? Label them.
Note: The ground node has been selected for you, so you don't need to label that, but you need to include it in your node count.
- Notice that elements 1 - 6 and the voltage source V_s have either the *voltage across* or the *current through* them not labeled. Label the missing *voltages across* or *currents through* for elements 1 - 6, and the voltage source V_s , so that they all follow **passive sign convention**.

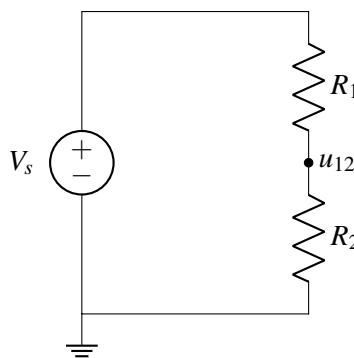
- (c) Express all element voltages (including the element voltage across the source, V_s) as a function of node voltages. This will depend on the node labeling you chose in part (a).
- (d) Write one KCL equation that involves the currents through elements 1 and 2.
- (e) Write a KVL equation for all the loops that contain the voltage source V_s . These equations should be a function of element voltages and the voltage source V_s .

3. Voltage divider

(Contributors: Adhyyan Narang, Panagiotis Zarkos, Sashank Krishnamurthy, Urmita Sikder)

In the following circuit, $V_s = 12\text{ V}$. Choose resistance values for R_1 and R_2 such that the current through each element is $\leq 0.8\text{ A}$ and $u_{12} = 6\text{ V}$. This is a **design problem**, so there can be more than one set of correct answers to this problem.

Hint: You can refer to [Note 11B](#) to solve this question. Other valid circuit solving methods are also acceptable.



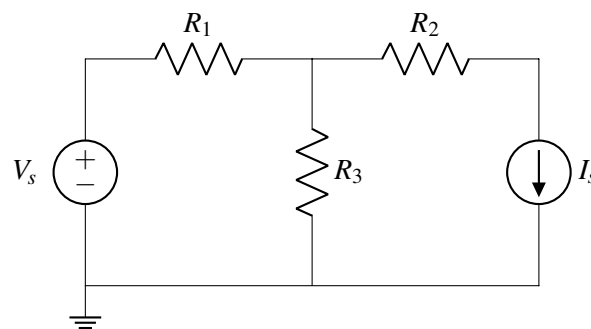
4. Circuit Analysis

(Contributors: Ava Tan, Aviral Pandey, Christos Adamopoulos, Matthew McPhail, Moses Won, Panos Zarkos, Raghav Anand, Titan Yuan, Urmita Sikder)

Learning Goal: This problem will help you practice circuit analysis using NVA method.

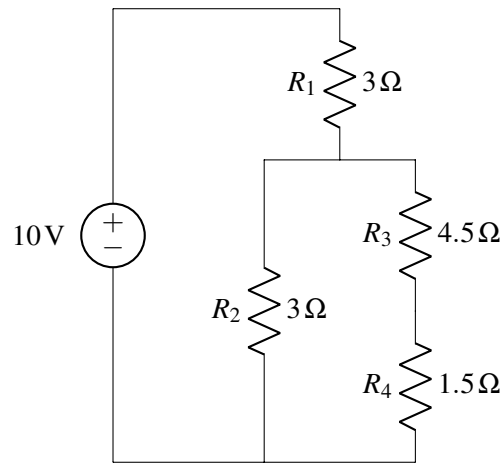
Using the steps outlined in lecture or in Note 11B, analyze the following circuit to calculate the currents through each element and the voltages at each node. Use the ground node labelled for you. You may use a numerical tool such as IPython to solve the final system of linear equations.

$$V_s = 5\text{ V}, I_s = 2\text{ A}, R_1 = R_2 = 2\Omega, R_3 = 4\Omega$$



5. Mechanical Circuits

Find the voltages across and currents flowing through all of the resistors.



6. It's a Triforce! (Contributors: Ava Tan, Elena Herbold, Ryan Tsang, Taehwan Kim, Urmita Sikder, Wahid Rahman)

Learning Goal: This problem explores passive sign convention and nodal analysis in a slightly more complicated circuit.

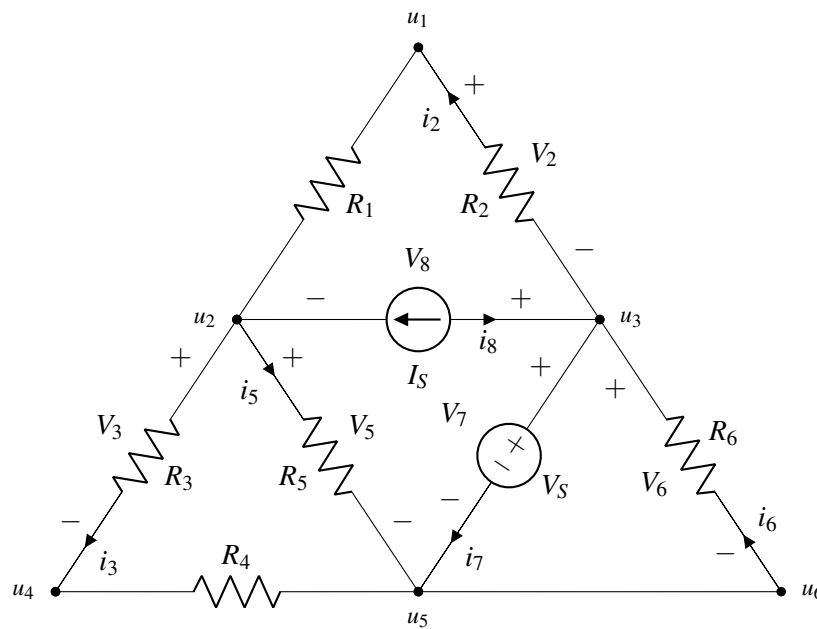


Figure 1: A triangular circuit consisting of a voltage source V_s , current source I_s , and resistors R_1 to R_6 .

- (a) Which of the elements I_s , V_s , R_2 , R_3 , R_5 , or R_6 in Figure 1 have current-voltage labeling that violates Passive Sign Convention? There could be more than one possible element which violates Passive Sign Convention. Explain your reasoning.

- (b) In Figure 1, the nodes are labeled with $u_1, u_2, \dots$ etc. There is a subset of u_i 's in the given circuit that are redundant, i.e. there might be more than one label for the same node. Which node(s)? Justify your answer.
- (c) Redraw the circuit diagram by correctly labeling all the element voltages and element currents according to passive sign convention. (The component labels that were violating Passive Sign Convention in part (a), should be corrected by swapping the element voltage polarity. Also, the elements that have not been labeled yet, should be labeled following Passive Sign Convention.)
- (d) Write an equation to describe the current-voltage relationship for element R_4 in terms of the relevant i 's, R 's, and node voltages in this circuit.
- (e) Write the KCL equation for node u_2 in terms of the node voltages and other circuit elements.

7. Fred's Fruits

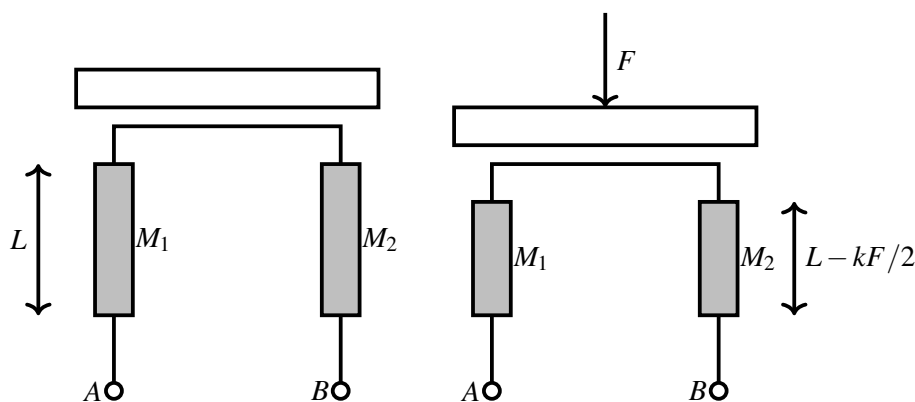
(Contributors: Ava Tan, Christos Adamopoulos, Matthew McPahil, Moses Won, Panos Zarkos, Urmita Sikder, Wahid Rahman)

Learning Goal: This problem will introduce the process of designing a sensing circuit for the purpose of measuring a physical quantity. This will also help to build your intuition for modeling physical elements.

Fred just got back from Berkeley Bowl with a bunch of mangoes, pineapples, and coconuts. He wants to sort his mangoes in order of weight, so he decides to use his knowledge from EECS16A to build a scale.

He finds two identical bars of material (M_1 and M_2) of length L (in meters) and a cross-sectional area (i.e. width $\times$ thickness) of A_{ca} (in meters²). The bars are made of a material with resistivity ρ . He knows that the **length of these bars decreases** by k meters per Newton of force applied, while the **cross-sectional area remains constant**.

He builds his scale as shown below, where the top of the vertical bars are connected with an ideal electrical wire. The left side of the diagram shows the scale at rest (with no object placed on it), and the right side shows it when the applied force is F (Newtons). The force F is equally distributed between two bars, causing the length of each bar to decrease by $kF/2$ meters. Fred's mangoes are not very heavy, so the change in each bar's length is very small compared to the total length (mathematically, we can write this by using the "much smaller than" symbol $\ll$, i.e. $kF/2 \ll L$).



- (a) Let R_{AB} be the resistance between nodes A and B with the weights on the scale. Write an expression for R_{AB} as a function of A_{ca} , L , ρ , F , and k . *Hint: You can start by representing each bar as a resistor; then find how they are connected.*

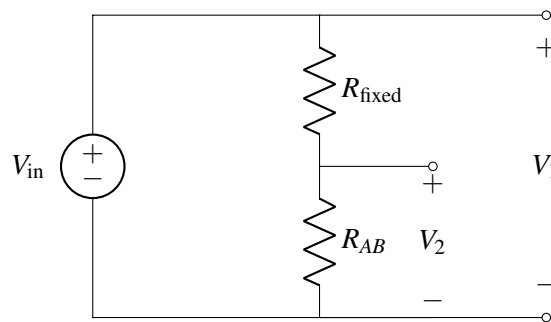
- (b) Fred wants to measure a voltage that changes based on how much weight is placed on his scale. He knows that R_{AB} will change with the weight on the scale.

Design a circuit for Fred that outputs a voltage that is some function of the weight F . **Your circuit should include R_{AB} , and you may use any number of voltage sources and resistors in your design.** Be sure to **label** where the voltage should be measured in your circuit.

Also provide an **expression** relating the output voltage of your circuit to the force applied on the scale. This expression can contain any necessary parameters.

Hint: If you connected only a voltage source across A and B and measured the voltage (V_{AB}) between A and B, would V_{AB} change based on the value of R_{AB} ? It turns out it wouldn't. Why?

Hint: Consider the following circuit, where R_{fixed} is a value you know and choose. Which voltage would you choose to measure in the circuit to help you determine the weight on the scale: V_1 or V_2 ?



8. Homework Process and Study Group

Who did you work with on this homework? List names and student ID's. (In case you met people at homework party or in office hours, you can also just describe the group.) How did you work on this homework? If you worked in your study group, explain what role each student played for the meetings this week.